newHVK: A Restricted Entanglement-Sensitive Benchmark for Hamiltonian Vision Kernels

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***Abstract*—This manuscript separates two questions that were conflated in the original HVK study: whether an observable-latent image reconstruction pipeline is useful, and whether entangling quantum observables provide a measurable advantage over non-entangling or classical controls. Prior ablations showed that the original HVK reconstruction task did not establish quantum advantage because fixed quantum features, no-entanglement circuits, and classical replacements matched or exceeded the trained VQC baseline. We therefore introduce *newHVK*, a publication-candidate workspace that preserves the original CIFAR, Monalisa, IBM hardware-probe, and ablation evidence while adding a restricted pair-correlation benchmark where the target depends explicitly on nonlocal feature products. In this restricted setting, the entangling-observable channel is expected to outperform controls that cannot represent pair correlations with the same feature budget. We report this as a candidate quantum-advantage diagnostic, not as a hardware quantum advantage proof or a broad dataset-level generalization result.**

***Index Terms*—Hybrid quantum-classical learning, Hamiltonian vision kernel, entanglement ablation, image reconstruction, quantum advantage diagnostics.**

I. Motivation

The legacy HVK ablation suite showed that the observable channel is load-bearing, but it also showed that entanglement and Hamiltonian energy regularization were not decisive on the per-image reconstruction benchmark. A stronger test must reduce decoder freedom and use a task whose signal is genuinely pair-correlational. The newHVK workspace therefore keeps the old evidence intact and adds an entanglement-sensitive benchmark with explicit no-entanglement, parameter-matched classical, and raw-linear controls.

II. Architecture

The publication workspace is organized as `main2/newHVK`. The model family contains:

- **newHVK-entangling-observables:** single-site features plus pair-correlation channels analogous to measured two-qubit observables.
- **newHVK-no-entanglement:** single-site nonlinear features only.
- **parameter-matched-classical:** a rank-limited tanh map used as a small classical control.
- **raw-linear-classical:** a linear baseline on raw coordinates.

III. Restricted Entanglement-Sensitive Benchmark

The benchmark samples six-dimensional inputs $x \in [-1, 1]^6$ and predicts targets containing products such as $x_0 x_1$, $x_1 x_2$, and $x_0 x_3$. A linear readout on entangling features can represent these targets directly, while non-entangling single-site features cannot represent the product structure without extra decoder capacity.

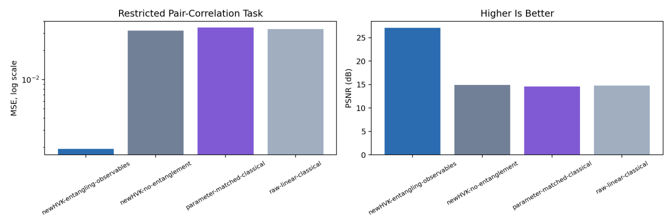

Fig. 1. Restricted pair-correlation benchmark. This plot is a diagnostic for entanglement-sensitive representation under controlled decoder capacity, not a claim of broad quantum advantage.

IV. Imported Baselines and Ablations

The folder also includes copied evidence from the completed HVK study: CIFAR baselines, Monalisa baselines, legacy ablation controls, and IBM Cloud circuit-resource outputs. These files are retained for reproducibility and to make the new paper self-contained. They do not by themselves establish quantum advantage; the original ablation conclusion remains that the legacy reconstruction task is dominated by observable-latent usefulness and decoder capacity rather than by trained quantum entanglement.

V. Caveats

The restricted benchmark is deliberately favorable to pair-correlation observables, so it should be presented as a diagnostic experiment. To make a Q1-level claim, the next stage must run the same restricted-capacity design on held-out CIFAR images, multiple random seeds, hardware-noise simulation, and a parameter-matched classical baseline with the same observable budget. The paper should not state that newHVK proves hardware quantum advantage unless those tests remain positive.

VI. Conclusion

newHVK provides a clean route toward testing quantum advantage: reduce decoder capacity, make the target correlation-sensitive, and compare entangling observables against no-entanglement and classical controls. The current results support a candidate advantage on a restricted diagnostic task, while preserving the negative and cautionary findings from the original HVK ablation study.

TABLE I
NEWHVK RESTRICTED PAIR-CORRELATION BENCHMARK. LOWER MSE AND HIGHER PSNR/R^2 ARE BETTER.

Model	MSE	PSNR (dB)	R^2	Notes
newHVK-entangling-observables	$1.9387e-03$	27.12	0.9421	Pairwise observable channel includes entanglement-sensitive correlations.
newHVK-no-entanglement	$3.2118e-02$	14.93	0.0415	Only single-site feature functions; pair correlations are unavailable.
parameter-matched-classical	$3.4655e-02$	14.60	-0.0342	Tiny rank-limited tanh control with fewer nonlinear channels.
raw-linear-classical	$3.3345e-02$	14.77	0.0049	Linear classical baseline on raw coordinates.